Quantum Hardware Design and Optimization
Laboratory 1 – Exercise 1 (optional circuit)

Group 4

Evaluation by hand of the state evolution

Two-qubit states are written as $|q_0 q_1\rangle$, with the basis ordered as $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. Layers 0–4 are identical to the ones of the main circuit.

Layers 0–1 – initial state and $H \otimes H$

$$|\psi_{t_0}\rangle = |00\rangle, \qquad |\psi_{t_1}\rangle = (H \otimes H)\,|\psi_{t_0}\rangle = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \;\rightarrow\; |{+}{+}\rangle\,.$$

Layer 2 – CZ

$$|\psi_{t_2}\rangle = CZ\;|\psi_{t_1}\rangle = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2}\big(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,\big).$$

Layer 3 – $H \otimes H$

$$|\psi_{t_3}\rangle = (H{\otimes}H)\,|\psi_{t_2}\rangle = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 2 \\ 2 \\ 2 \\ -2 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2}\big(\,|00\rangle{+}|01\rangle{+}|10\rangle{-}|11\rangle\,\big).$$

Layer 4 – $X \otimes X$

$$X \otimes X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

$$|\psi_{t_4}\rangle = (X \otimes X)\,|\psi_{t_3}\rangle = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2}\big(\,-|00\rangle + |01\rangle + |10\rangle + |11\rangle\,\big).$$

Layer 5 – $\mathbb{I} \otimes H$

$$\mathbb{I} \otimes H = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix},$$

$$|\psi_{t_5}\rangle = (\mathbb{I}\otimes H)\,|\psi_{t_4}\rangle = \frac{1}{2\sqrt{2}}\begin{pmatrix}1&1&0&0\\1&-1&0&0\\0&0&1&1\\0&0&1&-1\end{pmatrix}\begin{pmatrix}-1\\1\\1\\1\end{pmatrix} = \frac{1}{2\sqrt{2}}\begin{pmatrix}0\\-2\\2\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\-1\\1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\big(-|01\rangle+|10\rangle\big).$$

Layer 6 – $CNOT$ (control q_0, target q_1)

$$CNOT = |0\rangle\langle 0|\otimes\mathbb{I} + |1\rangle\langle 1|\otimes X = \begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix} + \begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix} = \begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix},$$

$$|\psi_{t_6}\rangle = CNOT\,|\psi_{t_5}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}\begin{pmatrix}0\\-1\\1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\-1\\0\\1\end{pmatrix} = \frac{1}{\sqrt{2}}\big(-|01\rangle+|11\rangle\big).$$

Layer 7 – $\mathbb{I}\otimes H$

$$|\psi_{t_7}\rangle = (\mathbb{I}\otimes H)\,|\psi_{t_6}\rangle = \frac{1}{2}\begin{pmatrix}1&1&0&0\\1&-1&0&0\\0&0&1&1\\0&0&1&-1\end{pmatrix}\begin{pmatrix}0\\-1\\0\\1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}-1\\1\\1\\-1\end{pmatrix} = \frac{1}{2}\big(-|00\rangle+|01\rangle+|10\rangle-|11\rangle\big).$$

Layer 8 – $X\otimes X$

$$|\psi_{t_8}\rangle = (X\otimes X)\,|\psi_{t_7}\rangle = \frac{1}{2}\begin{pmatrix}0&0&0&1\\0&0&1&0\\0&1&0&0\\1&0&0&0\end{pmatrix}\begin{pmatrix}-1\\1\\1\\-1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}-1\\1\\1\\-1\end{pmatrix} = \frac{1}{2}\big(-|00\rangle+|01\rangle+|10\rangle-|11\rangle\big).$$

Layer 9 – $H\otimes H$

$$|\psi_{t_9}\rangle = (H\otimes H)\,|\psi_{t_8}\rangle = \frac{1}{4}\begin{pmatrix}1&1&1&1\\1&-1&1&-1\\1&1&-1&-1\\1&-1&-1&1\end{pmatrix}\begin{pmatrix}-1\\1\\1\\-1\end{pmatrix} = \frac{1}{4}\begin{pmatrix}0\\0\\0\\-4\end{pmatrix} = \begin{pmatrix}0\\0\\0\\-1\end{pmatrix} = -|11\rangle.$$

Differences with the main circuit

The state $|\psi_{t_7}\rangle$ obtained here coincides with $|\psi_{t_5}\rangle$ of the main circuit, and the final state is again $-|11\rangle$. This is because the sequence of layers 5–7 is an equivalent implementation of the CZ gate:

$$(\mathbb{I}\otimes H)\,CNOT\,(\mathbb{I}\otimes H) = CZ,$$

since $HXH = Z$ on the target qubit. The optional circuit is therefore the same Grover iteration, with the CZ of the diffuser decomposed into H and $CNOT$ gates; the price is a deeper circuit (two additional layers).